D1: Baseline reproduction of the OxygenModel published results

Can a third party regenerate Cakin et al. 2026 from the deposited data today?

OxygenModel D-series

2026-07-26

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Verdict

Yes — with two named caveats. A third party who clones this repository today, runs `Rscript scripts/00_install.R` and then `bash scripts/run_all.sh`, regenerates **19 of the 22 committed tables that any script in the repository produces**, and of those 19, **10 agree to exactly 0.0** — bit-for-bit, not merely within tolerance. The whole run takes **63 seconds** and needs no interaction. The two caveats are:
(1) Figure 6 is produced by no script in this repository. Six committed tables, three committed text files and five committed TIFFs belong to Fig 6 (r vs R, the pooled / between-taxon / within-taxon mixed models). The modular pipeline does not make them, and neither does `scripts/original_scripts/OxygenModel.R` — which writes three *differently named* Fig 6 tables whose numbers do not match the committed ones. The script that produced the published Figure 6 is not in the repository.
(2) `results/` did not come from `scripts/original_scripts/`. The committed `results/tables/` were produced by the **modular** pipeline: they carry columns only the modular code writes (`fit_start_min`, `curve_code`, `O2_fit`, `cell_carbon_fg`), and they match this run of the modular pipeline to 0.0 while differing from a fresh run of the original scripts by up to 198%. So D1's load-bearing check does not have the meaning it was set up to have: it confirms the modular pipeline is self-consistent, not that it preserves numbers the original scripts produced. On the bacterial O_2 series the two implementations **do not agree** — growth rate r differs by up to 0.63% and the scaling parameter K by up to 23%, because they fit different windows (§3.3). On the temperature/CUE experiment and the synthetic-recovery validation they agree to **exactly 0.0**.
Nothing in the analysis was changed to reach this result. Three things had to change to make the pipeline *execute* unattended; all three are listed in §6.1.

1 What was done

`icakin/OxygenModel` is the reference implementation of the oxygen-dynamics method published as Cakin et al. 2026 (Cakin et al. 2026). At commit `0db2697` the repository had just been restructured from four monolithic scripts into a modular pipeline (`config.R`, `01`–`10`, `run_all.R`), with the pre-restructure code retained under `scripts/original_scripts/`. The restructure had never been checked against the published outputs.

D1 pinned the environment, made the pipeline run unattended, regenerated everything into a parallel output tree, and compared three trees:

Tree	What it is	Where
COMMITTED	the outputs in git	`results/tables`, `results/figures`
MODULAR	this run of `config.R` + `01`–`10`	`runs/D1_baseline`
ORIGINAL	this run of the five pre-restructure scripts	`runs/D1_originals`

The dataset is 15 bacterial taxa × 5 replicates = 75 dissolved-O_2 series, plus a separate *Pseudomonas* temperature gradient (6 temperatures × 4 replicates). All fits use the same normalised model, with N_0 fixed to 1 inside the fit:

$$O_2^{\mathrm{norm}}(t) = O_{2,0} + \frac{K}{r}\left(1 - e^{rt}\right)$$

fitted by `nlsLM` (Elzhov et al. 2023), with per-cell respiration reconstructed afterwards as $R = K \cdot O_{2,\mathrm{ref}}/N_0$.

Non-destructive. `results/` and `data/` were verified **byte-identical** before and after the run by SHA-256 over all 57 files (`env/pre_run_checksums.txt`), not by assertion.

1.1 Wall time

The full pipeline, unattended, run `20260726_211928`:

Table 2: Per-stage wall time. 03 and 04 are the Shiny apps: they are sourced as a parse-and-inputs check, never launched.

Stage	Script	Mode	Seconds	Status
01_longdata	01_longdata.R	run	1.95	ok
02_trimming	02_trimming.R	run	8.31	ok
03_trim_selector	03_trim_selector.R	source	2.21	ok
04_experiment_inputs	04_experiment_inputs.R	source	1.75	ok
05_oxygen_fits	05_oxygen_fits.R	run	7.03	ok
06_main_figures	06_main_figures.R	run	7.93	ok
07_cutoff_sensitivity	07_cutoff_sensitivity.R	run	10.04	ok
08_montecarlo_N0	08_montecarlo_N0.R	run	3.33	ok
09_simulation_recovery	09_simulation_recovery.R	run	15.71	ok
10_temperature_cue	10_temperature_cue.R	run	4.78	ok
TOTAL			63.04	

For reference, the five original scripts in the scratch tree took 49 s in total (OxygenModel.R 16.9 s; MC.R 0.7 s; 05CUTOFF.R 12.8 s; Simulation.R 14.8 s; CUE.R 3.4 s).

All ten stages ran. **Nothing was skipped**, silently or otherwise: `Skipped_Series_Log.csv` is empty, and 02 reports *kept 75 curves; skipped 0.* The only warnings in the whole log are two `nls.lm resetting 'maxiter' to 1024` notices from the temperature-response fits in stage 10.

2 Comparison D1 — modular pipeline vs the committed outputs

Classification: **IDENTICAL** (max relative difference $< 10^{-10}$), **NUMERICALLY EQUIVALENT** ($< 10^{-6}$), **DIVERGENT**, **NOT REGENERATED**. Rows are aligned on their natural key columns; `manual_fit_windows.csv` and `plot_exclude_points.csv` are excluded because they are manual curation and are *inputs* to the run.

19 IDENTICAL, 3 DIVERGENT, 6 NOT REGENERATED.

The 6 NOT REGENERATED tables are all Fig 6: `data_Fig6_pooled.csv`, `data_Fig6_RIS_COLLAPSED_centeredX` `Fig6_between_taxon_means.csv`, `Fig6_per_taxon_slopes.csv`, `Fig6_taxon_specific_lines_COLLAPSED_cen` `Fig6b_taxon_specific_slopes.csv`. Three committed text files are in the same position (`Fig6_pooled_regression.txt`, `Fig6_within_taxon_RIS.txt`, `mixed_model_Fig6_RIS_centeredX.txt`).

Table 3: D1: every committed CSV, compared with the modular pipeline's regenerated copy. The worst-disagreeing column for each table is in D1_modular_vs_committed_columns.csv.

Table	Class	Rows	Max abs diff	Max rel diff
data_Fig6_pooled.csv	NOT REGENERATED	75	–	–
data_Fig6_RIS_COLLAPSED_centere...	NOT REGENERATED	75	–	–
Fig6_between_taxon_means.csv	NOT REGENERATED	15	–	–
Fig6_per_taxon_slopes.csv	NOT REGENERATED	15	–	–
Fig6_taxon_specific_lines_COLLA...	NOT REGENERATED	1800	–	–
Fig6b_taxon_specific_slopes.csv	NOT REGENERATED	15	–	–
growth_rates_combined.csv	IDENTICAL	75	0.00e+00	0.00e+00
method_effects_estimates.csv	IDENTICAL	3	1.26e-14	6.92e-13
method_effects_significance.csv	DIVERGENT	3	6.71e-05	2.68e-04
N0_MC_ourmodel_overall_summary_...	IDENTICAL	4	4.44e-16	9.97e-16
N0_MC_ourmodel_taxon_summary_AL...	IDENTICAL	60	4.44e-16	1.13e-15
Oxygen_All_Long.csv	IDENTICAL	15450	0.00e+00	0.00e+00
Oxygen_Curve_Code_Key.csv	IDENTICAL	75	–	–
Oxygen_Data_Filtered.csv	IDENTICAL	9462	4.44e-15	9.38e-16
Oxygen_Data_Smoothed_Trimmed.csv	IDENTICAL	9462	4.44e-15	9.38e-16
oxygen_fit_curves.csv	IDENTICAL	8035	0.00e+00	0.00e+00
oxygen_fit_results.csv	IDENTICAL	75	0.00e+00	0.00e+00
oxygen_model_comparison_O2_ge_0...	IDENTICAL	1	0.00e+00	0.00e+00
oxygen_model_results_good_only_...	IDENTICAL	24	0.00e+00	0.00e+00
oxygen_model_results_main.csv	IDENTICAL	75	0.00e+00	0.00e+00
oxygen_model_results_O2_ge_0.5.csv	IDENTICAL	75	0.00e+00	0.00e+00
oxygen_results_with_R.csv	IDENTICAL	75	0.00e+00	0.00e+00
Oxygen_Trimmed_Series_Metadata.csv	IDENTICAL	75	4.55e-13	8.73e-16
SharpeSchoolfield_Temperature_P...	IDENTICAL	3	1.78e-14	4.82e-15
Skipped_Series_Log.csv	IDENTICAL	0	–	–
synthetic_parameter_recovery_re...	DIVERGENT	12150	5.34e-08	1.15e-05
Table_S2_synthetic_parameter_re...	DIVERGENT	27	1.27e-08	9.09e-06
Table_S3_oxygen_model_compariso...	IDENTICAL	75	0.00e+00	0.00e+00

2.1 The three DIVERGENT tables, in detail

Table 4: D1: the only three tables that do not reproduce.

Table	Max rel diff	Worst column
method_effects_significance.csv	2.684e-04	p.value
synthetic_parameter_recovery_results.csv	1.145e-05	r_est
Table_S2_synthetic_parameter_recovery_summary.csv	9.086e-06	R_rel_bias

- **`method_effects_significance.csv`** — worst disagreement 2.684e-04 relative, on `p.value`. These are `multcomp::glht` Tukey p-values, computed by simulation from a multivariate normal; they are not deterministic between runs. All three comparisons are non-significant in both trees, so nothing the paper claims turns on it.
- **`synthetic_parameter_recovery_results.csv`** and **`Table_S2_synthetic_parameter_recovery_summary.csv`** — worst disagreement 1.145e-05 relative. This one is informative: the modular `09_simulation_recovery.R` and the original `Simulation.R` agree with each other to **exactly 0.0** (§3.2), and *both* differ from the committed file by roughly 10^{-5}. The committed synthetic tables were therefore produced by an earlier run, under a different R or `minpack.lm` version. The bias and RMSE conclusions are unaffected at that magnitude.

Is the pipeline itself deterministic? Yes, apart from that one file. The whole pipeline was run twice, from a clean output tree both times; every regenerated table came back **byte-identical between the two runs except `method_effects_significance.csv`**, whose `glht` p-values moved in the fourth decimal place. So the 2 remaining DIVERGENT results are genuine differences from the committed files, not run-to-run noise.

The two committed mixed-model text summaries reproduce with identical numbers:

Table 5: D1: committed .txt model summaries. The two that differ do so only because a newer lmerTest wraps the first line, which shifts every subsequent line.

File	Class	Lines (committed / modular)	Note
Fig6_pooled_regression.txt	NOT REGENERATED	–	no counterpart in modular
Fig6_within_taxon_RIS.txt	NOT REGENERATED	–	no counterpart in modular
mixed_model_FC_summary.txt	NUMERICALLY EQUIVALENT (formatting differs)	26 / 27	33 numeric tokens, max relative difference 0
mixed_model_Fig6_RIS_centeredX.txt	NOT REGENERATED	–	no counterpart in modular
mixed_model_OD600_summary.txt	NUMERICALLY EQUIVALENT (formatting differs)	26 / 27	36 numeric tokens, max relative difference 0

2.2 Figures

TIFFs will never be byte-identical, so figures are compared by whether the pipeline produces them at all; where it does, the underlying data are the tables above.

Table 6: D1: the committed figures. Five Fig 6 panels are produced by neither the modular pipeline nor the original scripts; the sixth name, Fig_6_COMBINED.tiff, is emitted by the original run but from different numbers.

Figure	Modular run	Original run	Verdict
Fig_2_oxygen_dynamics_facet4_no_overflow.tiff	yes	yes	regenerated
Fig_3_growth_rate_comparison.tiff	yes	yes	regenerated
Fig_4_growth_rate_regression_normalized_com...	yes	yes	regenerated
Fig_5_BlandAltman_AllReplicates_LME.tiff	yes	yes	regenerated
Fig_6_COMBINED.tiff	NO	yes	NOT regenerated
Fig_6_r_vs_R_pooled_MAIN.tiff	NO	no	NOT regenerated
Fig_6b_between_taxon.tiff	NO	no	NOT regenerated
Fig_6c_within_taxon.tiff	NO	no	NOT regenerated
Fig_6d_per_taxon_replicates.tiff	NO	no	NOT regenerated
method_effects_CI.tiff	yes	yes	regenerated
supp_Fig_3_oxygen_all_replicates_NORMALISED...	yes	yes	regenerated
Supp_Fig_S1_residuals.tiff	yes	yes	regenerated

7 of the 12 committed figures are regenerated. The 5 that are not are all Fig 6. `Fig_6_COMBINED.tiff` is the one case where the original scripts do emit a file of the same name — but `OxygenModel.R` writes it as `Fig_6_r_vs_R_RIS_derand_full_MAIN.tiff` plus a combined panel, and the committed Fig 6 *data* do not match what those scripts produce (§3.4).

The run also produces 13 figures that are not committed at all — the multi-page diagnostics (`oxygen_trimming_diagnostics.pdf`, `oxygen_model_fit_curves.pdf`), the Monte Carlo panels, Fig S2 and Fig 7.

3 Comparison D2 — modular pipeline vs the original scripts

3.1 Do the original scripts still run?

Yes, all five, unmodified. They were copied into a scratch tree (`runs/D1_originals`) and run from there; `scripts/original_scripts/` was never modified or executed in place. They write to `data/`, `plots/` and `Tables/` relative to the working directory, which the scratch tree provides.

One data-provisioning fix was needed, and only in the scratch tree: `OxygenModel.R` hard-codes `data/Ninoc_and_deltaTime_to_NO.csv`, while the deposited file is `data/Ninoc.csv`. The scratch tree supplies the same file under both names. **No script was edited.** The minimal change that would remove even this would be to accept either filename, exactly as `config.R` already does via `NINOC_CSV_LEGACY`.

3.2 Per quantity

This is the load-bearing table: not "does file X match" but "does the number the paper reports match".

The split is clean and it is not random:

Table 7: D2: modular pipeline vs the original scripts, per quantity, on identical inputs.

Quantity	n	Max abs diff	Max rel diff	Verdict
growth rate r (min^-1)	75	1.78e-04	6.33e-03	DIVERGENT
scaling K	75	3.60e-04	2.32e-01	DIVERGENT
fitted intercept O2_0	75	2.66e-03	2.65e-03	DIVERGENT
reference O2 (O2_ref)	75	9.00e-02	1.52e-02	DIVERGENT
fit window length T_end (min)	75	4.30e+01	3.28e-01	DIVERGENT
total O2 consumed C_tot	75	3.85e+00	6.94e-01	DIVERGENT
pseudo R^2	75	2.03e-04	2.03e-04	DIVERGENT
reconstructed N0 (cells/L)	75	1.12e+06	1.02e-02	DIVERGENT
per-cell respiration R	75	2.87e-10	2.29e-01	DIVERGENT
respiration R (fg C/cell/h)	75	6.43e+03	2.28e-01	DIVERGENT
growth G (fg C/cell/h)	75	1.13e+02	4.98e-01	DIVERGENT
r vs OD600 growth rate	75	0.00e+00	0.00e+00	IDENTICAL
r vs flow cytometry	75	0.00e+00	0.00e+00	IDENTICAL
r from O2	75	1.78e-04	6.33e-03	DIVERGENT
cutoff sensitivity r (main)	75	7.06e-05	6.80e-03	DIVERGENT
cutoff sensitivity r (O2>=0.5)	75	6.86e-05	6.80e-03	DIVERGENT
MC relative SD of R (median)	4	2.78e-16	6.96e-16	IDENTICAL
CUE at each temperature	24	0.00e+00	0.00e+00	IDENTICAL
growth C flux vs temperature	24	0.00e+00	0.00e+00	IDENTICAL
respiration C flux vs temp	24	0.00e+00	0.00e+00	IDENTICAL
thermal Topt (deg C)	3	0.00e+00	0.00e+00	IDENTICAL
activation energy E (eV)	3	0.00e+00	0.00e+00	IDENTICAL
deactivation energy Eh (eV)	3	0.00e+00	0.00e+00	IDENTICAL
synthetic recovery r bias	27	0.00e+00	0.00e+00	IDENTICAL
synthetic recovery R bias	27	0.00e+00	0.00e+00	IDENTICAL

- **The temperature / CUE experiment reproduces exactly.** Every quantity from `10_temperature_cue.R` versus `CUE.R` — CUE per temperature, growth and respiration carbon fluxes, T_{opt}, E, E_h — differs by **0.0**. `CUE.R` was ported without changing a number.
- **The synthetic parameter recovery reproduces exactly.** `09_simulation_recovery.R` versus `Simulation.R`: **0.0** on all 12150 fits and all 27 summary rows.
- **The OD600 / flow-cytometry reference growth rates reproduce exactly** — they are read straight from `data/OD_r_FC_r.csv` by both.
- **The bacterial O_2 fits do not.** r differs by up to 0.63%, K by up to 23%, total O_2 consumed by up to 69%, and per-cell respiration R by up to 23%.

3.3 Why the bacterial fits differ

Not a porting bug — a different fit window, and a deliberate one.

	Modular (02 + 05)	Original (`OxygenModel.R`)
Start of window	spline peak, then the manual window from the 03 app	`which.max(raw Oxygen)`, then a derivative onset trim
End of window	manual window from the 03 app	second inflection between the steepest drop and the plateau
Manual curation	all 75 curves have a hand-set window	none available

All 75 curves carry a manual window in `results/tables/manual_fit_windows.csv`. Relative to 02's automatic trim those windows start a median +4.95 min later and end a median -11.95 min earlier, giving fit windows a median 86% of the automatic length. `T_end_min` differs between the two implementations by up to 43 minutes. Since K and C_{tot} are integrals over the window, a 14% shorter window moves them far more than it moves r — which is exactly the pattern in the table above (r 0.6%, K 23%, C_{tot} 69%).

3.4 Which implementation produced the committed results?

The modular one. Running the originals and comparing them to `results/tables/`:

Three independent lines of evidence:

1. `results/tables/oxygen_results_with_R.csv` contains `fit_start_min`, `N_inoc_cells_per_L` and `cell_carbon_fg`. Only the modular 05 writes those. `OxygenModel.R` writes `N_inoculation_cells_per_L` instead.
2. `results/tables/Oxygen_Data_Filtered.csv` contains `curve_code` and `O2_fit`. Only the modular 02 writes those.
3. `results/tables/data_Fig6_pooled.csv` holds $\log_{10}$ of per-cell respiration and growth. Those values match the **modular** `oxygen_results_with_R.csv` to 4.4×10^{-16} — machine precision — and differ from the original scripts' rerun by up to 0.11 in $\log_{10} R$ and 0.30 in $\log_{10} G$.

So the Fig 6 *inputs* were computed by the modular pipeline, but the Fig 6 *script* is absent. That is the single largest reproducibility gap in the repository.

Table 9: D2b: a fresh run of the ORIGINAL scripts against the committed results.

Table	Class	Max rel diff
data_Fig6_RIS_COLLAPSED_centeredX.csv	DIVERGENT	1.90e+00
Fig6_taxon_specific_lines_COLLAPSED_centere...	DIVERGENT	2.81e-01
Fig6b_taxon_specific_slopes.csv	DIVERGENT	1.98e+00
growth_rates_combined.csv	DIVERGENT	6.33e-03
method_effects_estimates.csv	DIVERGENT	7.45e-04
method_effects_significance.csv	DIVERGENT	4.57e-02
N0_MC_ourmodel_overall_summary_ALL.csv	IDENTICAL	9.97e-16
N0_MC_ourmodel_taxon_summary_ALL.csv	IDENTICAL	1.42e-15
Oxygen_Data_Filtered.csv	DIVERGENT	0.00e+00
oxygen_fit_curves.csv	DIVERGENT	1.00e+00
oxygen_fit_results.csv	DIVERGENT	6.94e-01
oxygen_model_comparison_O2_ge_0.5_summary.csv	DIVERGENT	1.03e-01
oxygen_model_results_good_only_NEWformula.csv	IDENTICAL	0.00e+00
oxygen_model_results_main.csv	DIVERGENT	6.87e-01
oxygen_model_results_O2_ge_0.5.csv	DIVERGENT	2.07e-01
oxygen_results_with_R.csv	DIVERGENT	6.94e-01
SharpeSchoolfield_Temperature_Params_NEWfor...	IDENTICAL	4.82e-15
Skipped_Series_Log.csv	DIVERGENT	–
synthetic_parameter_recovery_results.csv	DIVERGENT	1.15e-05
Table_S2_synthetic_parameter_recovery_summa...	DIVERGENT	9.09e-06
Table_S3_oxygen_model_comparison_O2_ge_0.5.csv	DIVERGENT	1.80e+00

Table 10: D3: r_O2 regressed on the reference method, with a taxon random intercept, exactly as 06_main_figures.R does it. R2 (mixed) is the squared correlation between fitted and observed values - the number annotated on Fig 4. R2 (OLS) is the fixed-effects-only value.

Tree	Comparison	n	Intercept	Slope	R2 (mixed)	R2 (OLS)
committed	r_O2 ~ r_OD600	75	0.00287	0.9100	0.9435	0.7230
committed	r_O2 ~ r_FC	75	0.00310	0.8848	0.9680	0.9414
modular	r_O2 ~ r_OD600	75	0.00287	0.9100	0.9435	0.7230
modular	r_O2 ~ r_FC	75	0.00310	0.8848	0.9680	0.9414
original	r_O2 ~ r_OD600	75	0.00284	0.9120	0.9429	0.7229
original	r_O2 ~ r_FC	75	0.00307	0.8866	0.9677	0.9407

Table 11: D3: Bland-Altman bias per method, in units of per-minute growth rate, computed as Fig 5 computes it.

Tree	Comparison	n	Bias	Bias (% of mean r)	Lower LoA	Upper LoA	% within
committed	O2 - OD600	75	0.000859	3.70	-0.00680	0.00852	92.0
committed	O2 - FC	75	0.000482	2.08	-0.00352	0.00448	97.3
modular	O2 - OD600	75	0.000859	3.70	-0.00680	0.00852	92.0
modular	O2 - FC	75	0.000482	2.08	-0.00352	0.00448	97.3
original	O2 - OD600	75	0.000875	3.77	-0.00679	0.00854	92.0
original	O2 - FC	75	0.000497	2.14	-0.00351	0.00451	97.3

4 Comparison D3 — the paper's headline numbers

4.1 Growth-rate agreement against OD_{600} and flow cytometry

The paper reports $R^2 > \mathbf{0.9}$ against both methods. That holds, on the statistic the figure actually annotates: $R^2 = \mathbf{0.944}$ against OD_{600} and $R^2 = \mathbf{0.968}$ against flow cytometry, with slopes 0.910 and 0.885 and intercepts 0.00287 and 0.00310 min^{-1}.

Worth stating plainly: that R^2 includes the taxon random intercept. The fixed-effects-only (OLS) R^2 against OD_{600} is **0.723**, not > 0.9. Against flow cytometry it is 0.941, which does clear 0.9 either way. The claim reproduces as the paper computes it; a reader who assumed a simple regression would get a different number for OD_{600}.

The original scripts give the same picture to three decimal places (slope 0.9120 vs 0.9100) — the window difference of §3.3 does not move this conclusion.

4.2 Bland–Altman

Bias is **+0.000859** $\mathbf{min^{-1}}$ against OD_{600} (3.7% of the mean O_2 growth rate) and **+0.000482** $\mathbf{min^{-1}}$ against flow cytometry (2.1%). Both are positive: the O_2 method reads slightly *faster* than either reference. 92% and 97.3% of replicates fall within the limits of agreement.

On **no taxon-specific bias**: per-taxon biases against flow cytometry span -2.4e-03 to 4.3e-03 min^{-1} and change sign across taxa (8 of 15 negative), consistent with the claim. Against OD_{600} they span -5.8e-03 to 1.0e-02, and one taxon stands out: *Pseudomonas* at +0.01026 min^{-1}, about 5× the mean absolute bias of the other 14. Stated as a report, not a re-analysis.

Table 13: D3: SharpeSchoolfield_Temperature_Params_NEWformula.csv. All three trees agree to 0.0.

Tree	Response	Best model	Topt (C)	E (eV)	E 95% CI	Eh (eV)	Eh 95% CI
committed	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872	[4.337, 5.406]
committed	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000	[-4.303, 16.303]
committed	carbon_use_efficiency	two	32.44	0.180	[-1.289, 1.649]	6.000	[3.872, 8.128]
modular	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872	[4.337, 5.406]
modular	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000	[-4.303, 16.303]
modular	carbon_use_efficiency	two	32.44	0.180	[-1.289, 1.649]	6.000	[3.872, 8.128]
original	growth_r_per_hr	one	33.98	0.633	[0.571, 0.694]	4.872	[4.337, 5.406]
original	resp_C_fg_per_hr	one	37.53	0.712	[0.589, 0.836]	6.000	[-4.303, 16.303]
original	carbon_use_efficiency	two	32.44	0.180	[-1.289, 1.649]	6.000	[3.872, 8.128]

Table 12: D3: per-taxon Bland-Altman bias, in units of per-minute growth rate, committed tree.

Taxon	Bias vs OD600	Bias vs FC
Acinetobacter	-0.00113	-0.00012
Aeromonas_A	+0.00175	+0.00129
Aeromonas_B	+0.00322	-0.00104
Arthrobacter	-0.00578	-0.00076
Bacillus	+0.00041	+0.00269
Burkholderia	-0.00249	+0.00243
Buttiauxella	-0.00177	-0.00012
Chromobacterium	+0.00285	-0.00003
Chryseobacterium	+0.00020	-0.00045
Flavobacterium_A	+0.00136	+0.00084
Flavobacterium_B	+0.00191	+0.00428
Herbaspirillum	+0.00018	+0.00052
Pseudomonas	+0.01026	+0.00053
Serratia	-0.00190	-0.00047
Yersinia	+0.00381	-0.00237

4.3 *Pseudomonas* thermal parameters

All three trees are **identical to 0.0**. `best_model = one` is the one-sided Sharpe–Schoolfield form (Schoolfield, Sharpe, and Magnuson 1981), selected by AIC against the two-sided form and a Boltzmann–Arrhenius.

Two of the confidence intervals are worth flagging as reported, not as fixed: E_h for respiration is 6.000 [-4.303, 16.303] eV — the estimate sits exactly on its lower bound of 6.0 and the CI covers negative values — and E for CUE is 0.180 [-1.289, 1.649] eV, a CI spanning zero. Both are consequences of fitting six temperatures.

4.4 Where the CUE optimum falls relative to the growth optimum

Table 15: PART E: every file in data/. 'Consumed by' is abbreviated; the full text is in E1_data_audit.csv.

File	Rows	Cols	Taxa	Series	Missing	Consumed by
Cell_Counts.csv	75	6	15	75	0	04_experiment_inputs.R ...
Ninoc.csv	75	4	15	75	0	05_oxygen_fits.R
OD_r_FC_r.csv	75	8	15	75	0	06_main_figures.R (Figs...
Oxygen_Data_Filtered_CUE.csv	3265	5	1	4	0	10_temperature_cue.R
Oxygen_Data_Long.csv	15450	4	15	75	0	01_longdata.R -> 02 -> ...
taxon_cell_sizes.csv	15	6	15	–	0	config::cell_carbon_of(...

Table 14: D3: observed per-temperature means, Pseudomonas, four replicates each.

Temperature (C)	n	Mean r (1/h)	Mean growth (fg C/h)	Mean resp (fg C/h)	Mean CUE
20	4	0.575	3.28e+07	2.24e+08	0.1280
24	4	0.819	5.62e+07	2.41e+08	0.1904
28	4	1.153	1.02e+08	4.79e+08	0.1762
32	4	1.457	1.61e+08	6.15e+08	0.2091
36	4	1.430	1.55e+08	8.3e+08	0.1579
40	4	0.473	2.52e+07	6.84e+08	0.0355

The paper says CUE peaks around the growth optimum, at roughly 30–35 °C. **It does.** Fitted optima: growth **33.98 °C**, CUE **32.44 °C** — CUE peaks **1.54 °C below** the growth optimum, and both sit inside 30–35 °C. Respiration peaks later, at **37.53 °C**, which is the mechanism: respiration keeps rising past the growth optimum, so CUE turns over slightly before growth does.

Model-free, the observed means agree: mean growth rate and mean CUE both peak in the **32 °C** bin (1.457 h^{-1} and 0.2091), while mean respiration keeps rising to 36 °C.

5 Data integrity and the QC gate

5.1 The audit

No missing values anywhere in `data/` — 0 NA cells across all six files. Units are consistent but entirely implicit: nothing in `data/` declares that `Time` is minutes, `Oxygen` is mg O_2 L^{-1} or `Temperature` is °C.

One file is unused. `data/Cell_Counts.csv` is read by no automated stage. Only the `04_experiment_inputs.R` Shiny app touches it, and only as a fallback list of taxon names when `Oxygen_All_Long.csv` is absent. Its `OD_Initial/OD_Final/FC_Initial/FC_Final` columns are duplicated — not identically — in `data/OD_r_FC_r.csv`, which *is* used, by `06`.

5.2 `FC_Initial` versus `N_inoculation_cells_per_L`

These are the same quantity from two sources: the flow-cytometry count at inoculation, and the inoculation density that anchors N_0 and therefore all absolute respiration. Taking the `04` app's own stated convention — counts are cells µL^{-1}, multiplied by 10^6 to give cells L^{-1} — they do not agree.

Table 16: PART E: ratio of N_inoculation_cells_per_L (data/Ninoc.csv) to FC_Initial x 1e6 (data/Cell_Counts.csv), per taxon. 'FC sources disagree' counts replicates where FC_Initial differs between Cell_Counts.csv and OD_r_FC_r.csv.

Taxon	n	FC sources disagree	Median ratio	Min	Max
Acinetobacter	5	5	0.356	0.180	0.541
Aeromonas_A	5	4	0.171	0.095	0.257
Aeromonas_B	5	5	0.087	0.049	0.148
Arthrobacter	5	5	0.162	0.104	0.189
Bacillus	5	4	0.058	0.037	0.112
Burkholderia	5	5	0.141	0.119	0.177
Buttiauxella	5	5	0.174	0.145	0.264
Chromobacterium	5	5	0.141	0.105	0.164
Chryseobacterium	5	5	0.311	0.262	0.414
Flavobacterium_A	5	4	0.134	0.098	0.295
Flavobacterium_B	5	5	0.183	0.168	0.335
Herbaspirillum	5	5	0.216	0.145	0.293
Pseudomonas	5	5	0.183	0.093	0.270
Serratia	5	5	0.084	0.064	0.193
Yersinia	5	5	0.055	0.025	0.073

Across all 75 replicates the ratio has median **0.151** and spans **0.025 to 0.541** — a 21-fold spread. That is not a constant unit offset; it is a per-replicate disagreement. The per-replicate ratios are in `E2_FC_vs_Ninoc_per_replicate.csv`.

Separately, `FC_Initial` itself differs between `data/Cell_Counts.csv` and `data/OD_r_FC_r.csv` in **72 of 75 replicates** (for example *Acinetobacter* R1: 94 vs 83). There are effectively three inoculum numbers in the repository for each series.

Reported only. D1 changes nothing here. But note the consequence: N_0 scales per-cell respiration R linearly, so a systematically low N_0 inflates R by the same factor.

5.3 The QC gate

Table 17: PART E: the automatic QC gate, all 75 series.

Criterion	Threshold	Series	Failures
REL_SE	max rel SE < 0.15	75	0
PVAL	max p < 0.001	75	0
R2	pseudo R2 >= 0.90	75	0
RESID	residual range < 0.20	75	0
K_BOUNDS	0 < K < 0.5	75	0
R_BOUNDS	1e-4 <= r <= 0.1	75	0
AIC	AIC gain >= 10	75	0
MAPE	MAPE < 0.15	75	0

All 75 of 75 series pass all eight criteria. Not narrowly, either: the tightest series still clears its binding threshold by 92% of the available margin. Typical values are pseudo-$R^2 > 0.9998$ against a 0.90 threshold, MAPE about 0.001 against 0.15, and AIC gains of 650–870 against a required 10.

Table 18: PART E: automatic gate versus manual curation.

Metric	Value
series in the run	75
series that produced a fit	75
series passing fit_ok (automatic gate)	75
series failing fit_ok (automatic gate)	0
series removed by plot_exclude_points.csv (manual)	0
overlap: manually excluded AND automatically failed	0
manually excluded but automatically PASSING	0
automatically failed but NOT manually excluded	0
curves with a manual fit window in manual_fit_windows.csv	75
of which both fit_start and fit_end are set	75
of which only one side is set	0

Is the automatic gate doing the work, or the manual curation? Neither is doing any *filtering*: 0 series fail the gate and 0 are manually excluded (`plot_exclude_points.csv` is empty), so the overlap question is vacuous — there is nothing to overlap.

But that is not the same as saying manual curation is idle. The curation that *is* load-bearing is `manual_fit_windows.csv`, which sets an explicit window on **all 75 curves, both sides on every one**. Those windows, not the automatic trimmer, determine every fitted number, and they are why the modular pipeline and the original scripts disagree (§3.3). The QC gate is a formality on this dataset; the hand-set windows are the analysis decision.

5.4 `NO_BACKPROJECT` and the amplification factor

`config.R` sets `NO_BACKPROJECT <- TRUE`, so $N_0 = N_{\text{inoc}} \cdot e^{r\delta}$, with δ the inoculation $\rightarrow$ O_2-start delay.

Table 19: PART E: the N0 back-projection, all 75 series.

Statistic	Value
n series	75.0000
delta min (min)	46.0300
delta median (min)	60.0700
delta max (min)	81.0700
exp(r*delta) min	1.8046
exp(r*delta) median	3.8572
exp(r*delta) max	12.1305
N0/N_inoc median	3.8572
fraction of series with exp(r*delta) > 2	0.9733
fraction with exp(r*delta) > 5	0.2400

δ spans **46.03 to 81.07 minutes** and the amplification $e^{r\delta}$ has median **3.86**, reaching **12.13**. 97% of series are amplified more than twofold and 24% more than fivefold. The amplification is exponential in the *fitted* r, so fit noise in r propagates into N_0 — and hence into R — with a lever arm of δ, up to 81 minutes.

Table 20: PART E: back-projection amplification by taxon, ordered by median.

Taxon	n	delta median	exp(r d) min	exp(r d) median	exp(r d) max
Yersinia	5	55.07	6.76	8.44	12.13
Aeromonas_B	5	56.05	4.34	6.48	9.25
Chromobacterium	5	65.07	4.67	5.20	5.56
Bacillus	5	57.07	3.50	5.11	8.65
Arthrobacter	5	62.05	3.80	4.89	5.24
Flavobacterium_A	5	62.07	2.38	4.48	4.84
Serratia	5	54.07	3.15	4.18	4.81
Herbaspirillum	5	63.05	2.80	3.93	5.97
Pseudomonas	5	66.07	3.50	3.82	5.89
Burkholderia	5	63.05	3.29	3.74	5.12
Aeromonas_A	5	61.07	3.20	3.50	4.50
Buttiauxella	5	57.05	3.09	3.43	3.91
Chryseobacterium	5	72.07	2.57	3.30	3.49
Acinetobacter	5	58.05	1.81	2.45	2.70
Flavobacterium_B	5	59.05	1.80	2.26	2.42

Reported only. D2 investigates.

6 Blockers

6.1 What had to change to make the pipeline run

Three things, all of them execution blockers, none of them analysis decisions.

1. **`04_experiment_inputs.R` launched a browser in the non-interactive branch.** It read `if (interactive()) shinyApp(...) else runApp(..., launch.browser = TRUE)`, so any headless `source()` blocked forever waiting on a Shiny server. `03_trim_selector.R` ended in a bare top-level `shinyApp(ui, server)`. Both are now guarded behind `interactive() || OXYMODEL_LAUNCH_APPS=1`.
2. **`10_temperature_cue.R` was outside `run_all.R`'s loop.** The published *Pseudomonas* CUE result — Figure 7 and `SharpeSchoolfield_Temperature_Params_NEWformula.csv` — was regenerated by no automated path at all. It is now stage 10 of the run.
3. **`lmerTest` and `multcomp` were not installed on the machine.** `06_main_figures.R` loads both, so Figures 3/4/5 could not have been produced. They are now pinned in `renv.lock`.

Two documentation defects were corrected rather than left to mislead:

4. **`run_all.R` and `config.R` both claimed the OD600 / flow-cytometry comparison was "intentionally not included".** It is included: `06_main_figures.R` emits `Fig_3_growth_rate_comparison.tiff`, `Fig_4_growth_rate_regression_normalized_combined.tiff`, `Fig_5_BlandAltman_AllReplicates_LME.tiff` and `method_effects_CI.tiff` from

data/OD_r_FC_r.csv. The committed Figs 3–5 are regenerated by the modular pipeline and match it. What is genuinely absent is Fig 6.

5. **scripts/README.md claims 06_main_figures.R produces Fig 6.** It does not, and never did in this commit.

One switch was added, OXYMODEL_RESULTS, defaulting to results/ so default behaviour is unchanged. It redirects only outputs; the two app-written curation tables are read from the committed results/tables/ regardless, because they are inputs.

6.2 What still cannot be run

1. **Figure 6 cannot be regenerated by anything in this repository.** Six tables, three text files and five TIFFs have no producer. OxygenModel.R writes three differently-named Fig 6 tables whose numbers do not match the committed ones, and the committed data_Fig6_pooled.csv matches the *modular* fit results to machine precision — so the missing script consumed modular output. Recovering Figure 6 needs that script, or a re-derivation.
2. **OxygenModel.R cannot find its inoculum table as deposited.** It hard-codes data/Ninoc_and_deltaTime_to_N0.csv; the deposited file is data/Ninoc.csv. The scratch tree supplies both names. The minimal fix would be to accept either, as config.R already does.
3. **The committed synthetic-recovery tables cannot be reproduced exactly.** Both current implementations agree with each other to 0.0 and differ from the committed files by roughly 10^{-5} — an artefact of the R / minpack.lm version they were made with, now unrecoverable. Nothing the paper claims depends on the fifth decimal place of a bias estimate.
4. **data/Cell_Counts.csv cannot be reconciled with data/Ninoc.csv** from inside the repository (§5.2). It needs the experimentalists.

6.3 Correction to an earlier commit message

The commit D1: regenerate into runs/D1_baseline; three-way comparison states "21 of 22 regenerable tables IDENTICAL". The correct counts, from D1_modular_vs_committed_tables.csv, are **19 IDENTICAL, 3 DIVERGENT, 6 NOT REGENERATED**. The tables in §2 are authoritative.

Provenance

Table 21: Provenance. Environment fields come from env/versions.json, written by scripts/00_install.R.

Field	Value
Report	D1 — baseline reproduction
Git commit	0db2697a1da3841456a800eb26173edc89cd5912
Branch	main
Environment recorded (UTC)	2026-07-26 20:07:56 UTC
Pipeline run id	20260726_211928
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
OS	macOS Tahoe 26.2
renv lockfile	renv.lock — 135 packages, R 4.5.2
Quarto	1.8.27

Field	Value
PDF engine	lualatex — This is LuaHBTeX, Version 1.24.0 (TeX Live …
Output tree	runs/D1_baseline
Originals scratch tree	runs/D1_originals

Where every number in this report came from

Every table above is rendered directly from the file named here; no number was typed in by hand. All paths are relative to the project root.

Table 22: Number-to-source map.

Section	Source
§1.1 wall time	runs/D1_baseline/comparisons/D0_stage_timings.csv (from logs//stage_timings.csv)
§1.1 originals wall time	runs/D1_baseline/comparisons/D0_original_timings.csv
§1 byte-identical claim	env/pre_run_checksums.txt, verified with shasum -a 256 -c
§2 table classification	runs/D1_baseline/comparisons/D1_modular_vs_committed_tables.csv
§2 per-column detail	runs/D1_baseline/comparisons/D1_modular_vs_committed_columns.csv
§2 text summaries	runs/D1_baseline/comparisons/D1_modular_vs_committed_textfiles.csv
§2.2 figures	runs/D1_baseline/comparisons/D1_figure_inventory.csv
§3.2 per quantity	runs/D1_baseline/comparisons/D2_per_quantity.csv
§3 per table	runs/D1_baseline/comparisons/D2_modular_vs_original_tables.csv
§3.3 window shifts	runs/D1_baseline/comparisons/E3_manual_vs_auto_windows.csv
§3.4 original vs committed	runs/D1_baseline/comparisons/D2b_original_vs_committed_tables.csv
§4.1 R2 / slope / intercept	runs/D1_baseline/comparisons/D3_growth_rate_agreement.csv
§4.2 Bland-Altman	runs/D1_baseline/comparisons/D3_bland_altman.csv
§4.2 per-taxon bias	runs/D1_baseline/comparisons/D3_bland_altman_by_taxon.csv
§4.3 thermal parameters	runs/D1_baseline/comparisons/D3_thermal_parameters.csv
§4.4 CUE vs growth optimum	runs/D1_baseline/comparisons/D3_optima.csv, D3_cue_by_temperature.csv
§5.1 data audit	runs/D1_baseline/comparisons/E1_data_audit.csv
§5.2 FC vs Ninoc	runs/D1_baseline/comparisons/E2_FC_vs_Ninoc_by_taxon.csv, E2_FC_vs_Ninoc_per_replicate.csv
§5.3 QC gate	runs/D1_baseline/comparisons/E3_qc_gate_by_criterion.csv, E3_qc_gate_per_series.csv
§5.3 gate vs curation	runs/D1_baseline/comparisons/E3_gate_vs_manual_curation.csv
§5.4 back-projection	runs/D1_baseline/comparisons/E4_backprojection_overall.csv, E4_backprojection_by_taxon.csv
Provenance	env/versions.json, runs/D1_baseline/comparisons/D0_run_manifest.csv

The artefacts themselves are produced by `reports/D1_baseline/scripts/d1_compare.R` and `reports/D1_baseline/scripts/d1_audit.R`, from the trees built by `scripts/run_all.sh` and `reports/D1_baseline/scripts/run_originals.sh`. To regenerate everything in this report from a clean checkout, see `reports/RUNBOOK.md`.

References

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